Graph Transformations: Combinations

Questions 1–41

Question 1. Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = 3f(4x)$

b) $y = \frac{1}{2}f(\frac{1}{3}x)$

Question 2. Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = -f(x + 6)$

b) $y = f(-x) + 6$

Question 3. Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = \frac{1}{6}f(x-4)$

b) $y = f(8x) - 5$

Question 4. Prerequisite knowledge for Q4: Radians Describe a sequence of two transformations that will map $y = \sin x$ onto the image:

a) $y = 2\sin(x - \frac{\pi}{3})$

b) $y = \sin(\frac{1}{5}x) + 2$

Question 5. Describe a sequence of two transformations that will map $y = e^x$ onto the image:

a) $y = 5e^{x/3}$

b) $y = e^{-x} - 9$

Question 6. Describe a sequence of two transformations that will map $y = \frac{1}{x}$ onto the image:

a) $y = \frac{2}{x-4}$

b) $y = 5 - \frac{1}{x}$

Question 7.

a) Describe a sequence of two transformations that will map $y = \ln x$ onto the image $y = \ln(2x) - 3$.

b) Describe a **single** transformation that will map $y = \ln x$ onto the image $y = \ln(2x) - 3$.

Question 8. The point $P(-6, 12)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of P after $y = f(x)$ is mapped to each of these:

a) $y = 2f(3x)$

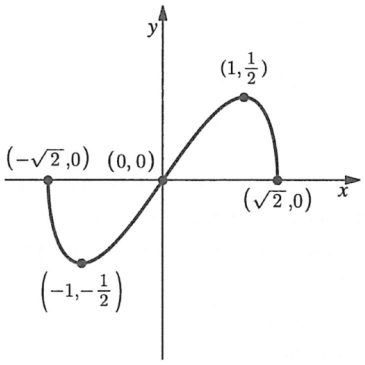
b) $y = -f(x - 5)$

Question 9. The point $Q(4, -3)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of Q after $y = f(x)$ is mapped to each of these:

a) $y = \frac{1}{3}f(\frac{1}{6}x)$

b) $y = f(-x) + 9$

Question 10. The graph of $y = f(x)$ is shown in the diagram below.



Sketch the graphs of

a) $y = 2f(\frac{1}{2}x)$

b) $y = -f(x + \sqrt{2})$

c) $y = f(2x) + \frac{1}{2}$

d) $y = f(-x) - 1$

Question 11. Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = 2f(x) + 3$

b) $y = 9f(x) - 8$

c) $y = \frac{1}{5}f(x) - 10$

d) $y = \frac{3}{2}f(x) + 20$

Question 12. Describe a sequence of two transformations that will map:

a) $y = 4f(x) - 1$ onto $y = 5f(x) + 2$

b) $y = -f(x) - 5$ onto $y = -\frac{1}{2}f(x) + 7$

Question 13. Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = f(3x + 1)$

b) $y = f(2x - 9)$

c) $y = f(2 - x)$

d) $y = f(-x - 3)$

Question 14. Describe a sequence of two transformations that will map:

a) $y = f(3x + 8)$ onto $y = f(6x - 4)$

b) $y = f(2 - x)$ onto $y = f(3 - \frac{1}{5}x)$

Question 15. The point $P(10, -3)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of P after $y = f(x)$ is mapped to each of these:

a) $y = 2f(x - 5) + 3$

b) $y = -f(x) - 9$

c) $y = f(3x + 5)$

d) $y = f(\frac{1}{4}x - 2)$

Question 16. Prerequisite knowledge for Q16: Radians Describe a sequence of two transformations that will map $y = \sin x$ onto the image:

a) $y = 4\sin x - 6$

b) $y = \sin(\frac{x}{6} - \frac{\pi}{9})$

c) $y = \cos 2x$

d) $y = 3\cos x$

Question 17. Describe a sequence of two transformations that will map $y = e^x$ onto the image:

a) $y = e^{6x-10}$

b) $y = e^{9-x}$

c) $y = 10e^x + 20$

d) $y = 18 - e^x$

Question 18. Describe a sequence of two transformations that will map $y = \frac{1}{x}$ onto the image:

a) $y = \frac{1}{5x-2}$

b) $y = \frac{8}{x} - 9$

c) $y = \frac{6-5x}{x}$

d) $y = \frac{2x+7}{20x}$

Question 19. Describe a sequence of two transformations that will map $y = \frac{1}{x^2}$ onto the image:

a) $y = -\frac{1}{5x^2}$

b) $y = 1 + \frac{2}{x^2}$

c) $y = \frac{2+3x^2}{6x^2}$

d) $y = \frac{1-6x^2}{8x^2}$

Question 20.

a) Describe a sequence of two transformations that will map $y = \ln(x + 1)$ onto the image $y = \ln(3x + 3)$.

b) Describe a **single** transformation that will map $y = \ln(x + 1)$ onto the image $y = \ln(3x + 3)$.

Question 21. Describe a sequence of two transformations that will map $y = 4x + 3$ onto the image:

a) $y = 8x + 7$

b) $y = 2x + 6$

Question 22. Describe a sequence of two transformations that will map $y = 4 - 5x$ onto the image:

a) $y = -10x - 8$

b) $y = 50x - 40$

Question 23. Write down the image of $y = 5x + 2$ after a translation by the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$, followed by a stretch, parallel to the x-axis, scale factor 2.

Question 24. Write down the image of $y = 8 - 6x$ after a stretch, parallel to the y-axis, scale factor $\frac{1}{3}$, followed by a translation by the vector $\begin{pmatrix} -\frac{2}{3} \\ \frac{4}{3} \end{pmatrix}$.

Question 25. Describe a sequence of two transformations that will map $y = x^2$ onto the image:

a) $y = 4x^2 + 9$

b) $y = 5 - x^2$

c) $y = \frac{1}{4}(x - 6)^2 + 3$

d) $y = 8(x + 1)^2 - 2$

e) $y = (4x + 1)^2$

f) $y = (\frac{1}{3}x - 9)^2$

g) $y = 2x^2 - 4x - 7$

h) $y = 3x^2 + 12x + 1$

i) $y = -(x + 1)(x - 9)$

j) $y = (6x - 4)(3x + 10)$

Question 26. Describe a sequence of two transformations that will map $y = x^2 + 2$ onto the image:

a) $y = 9x^2 + 10$

b) $y = 8x^2$

c) $y = 7 - x^2$

Question 27. Write down the image of $y = x^3$ after a stretch, parallel to the x-axis, scale factor $\frac{1}{2}$, followed by a translation by the vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

Question 28. Write down the image of $y = \ln x$ after a stretch, parallel to the x-axis, scale factor 9, followed by a translation by the vector $\begin{pmatrix} 6 \\ -\ln 3 \end{pmatrix}$, showing that it can be written in the form $y = \ln(ax + b)$, where a and b are rational numbers.

Question 29. Find the equation of the curve $y = x^2$ after a reflection in the line:

a) $y = 2$

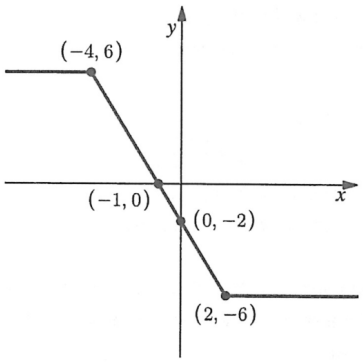
b) $x = 3$

Question 30. Prerequisite knowledge for Q30: Radians Find the equation of the curve $y = \cos x$ after a reflection in the line:

a) $y = -3$

b) $x = -\frac{\pi}{3}$

Question 31. The graph of $y = f(x)$ is shown in the diagram below.



Sketch the graphs of

a) $y = f(\frac{1}{2}x - 2)$

b) $y = f(2x + 5)$

c) $y = 3f(x) + 1$

d) $y = \frac{1}{2}f(x) - 4$

Question 32. Describe the transformation that maps $y = f(x)$ onto the image $x = f(y)$.

Question 33. Describe the transformation that maps $y = 8 - (x - 3)^2$ onto the image $x = 8 - (y - 3)^2$.

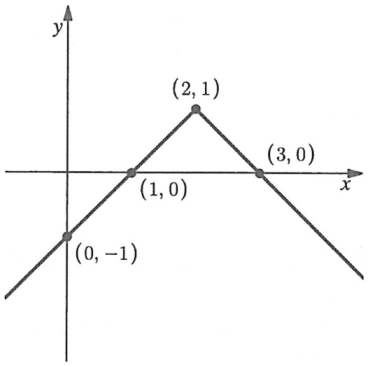
Question 34. Sketch the image of $(x - 1)^2 + y^2 = 9$ once it has been reflected in the line $y = x$.

Question 35. Describe a sequence of two transformations that will map $y = \ln x$ onto the image $y = e^{3x}$.

Question 36. Describe a sequence of two transformations that will map $y = \ln(x - 5)$ onto the image $y = e^{x}$.

Question 37. Describe a sequence of extbfthree transformations that will map $y = \ln x$ onto the image $y = 2e^x - 7$.

Question 38. The graph of $y = f(x)$ is shown in the diagram below.



Sketch the graph of $x = f(y)$.

Question 39. Describe a sequence of extbffour transformations that will map $y = f(x)$ onto the image $y = 2f(\frac{1}{3}x - 9) - 3$.

Question 40. Describe a sequence of extbffour transformations that will map $y = f(x)$ onto the image $y = \frac{3}{5}f(\frac{2}{5}x + 4) + 2$.

Question 41. Find the equation of the curve $y = x^2$ after a reflection in the line $y = -x$.